



INTAN KAMILIA BINTI ROSLAN

Title : AI Engineer | Data Engineer
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SUMMARY

AI Engineer and Data Engineer with hands-on experience designing and deploying production Machine Learning and LLM systems (RAG, agentic workflows with LangGraph/LangChain, vLLM model serving, Flowise orchestration) alongside large-scale ETL pipelines and data warehouses (Hadoop, PostgreSQL, star-schema data marts). Skilled in vector database administration and natural language processing (NLP).

TECHNICAL SKILLS

Languages	Python, SQL, R, JavaScript, Java, C++
Machine Learning / NLP	RAG pipelines, hybrid retrieval (FAISS + BM25), agentic workflow orchestration (LangGraph, LangChain), LLM chatflow design & orchestration (Flowise), prompt engineering, LLM evaluation, natural language processing (NLP), vector databases (pgvector, Supabase), embedding models, OpenAI GPT API, YOLOv11 (computer vision), vLLM model serving & context-window optimization
Data Engineering	ETL/ELT pipeline design (KNIME, Apache NiFi), source-to-warehouse migration (MariaDB/MySQL to Hadoop/HDFS data lake to PostgreSQL data warehouse), star-schema data mart design, PostgreSQL (administration, performance tuning, connection pooling), data pipeline monitoring & automation
Infrastructure	Windows Server & Linux (Ubuntu) administration, SSH/remote server management, PowerShell scripting, task scheduling & automated job monitoring (watchdog systems)
Automation	n8n workflow automation, including AI agent orchestration (Gemini), RSS feed ingestion, Google Sheets integration, and automated email drafting
Visualization & BI	Tableau, Power BI, Jupyter Notebook
Databases	PostgreSQL, MySQL, MariaDB, MongoDB, Oracle
Other	HTML5, CSS, PHP, Git

PROFESSIONAL EXPERIENCE

Opensoft Technologies Sdn Bhd | AI Engineer | Data Engineer October 2024 - Present

MBPJ AI-Powered Chatbot System

- Installed and configured vLLM on a single NVIDIA L40S GPU (46GB VRAM) to serve Qwen2.5-14B-Instruct (chat) and Qwen3-Embedding-0.6B (embedding) with fp8 quantization, budgeting GPU memory utilization across both services to fit production hardware constraints.
- Administered and tuned production Flowise chatflows and pgvector document stores, resolving LLM context-window overflow issues by optimizing token budgets (maxTokens, topK) against a 16K context ceiling.
- Diagnosed and fixed a RAG retrieval quality issue by identifying a misconfigured web-scraper loader that had introduced 6,000+ noise chunks into a document store, trimming the store down to relevant content and restoring answer accuracy.

SWCORP Enterprise Data Pipeline Management (Data Raya)

- Built and maintained ETL/ELT pipelines in KNIME sourcing multi-department data from MariaDB/MySQL systems into a 3-node Hadoop/HDFS cluster, then into a PostgreSQL data warehouse, using HDFS Connector and SSH-based ingestion routes.
- Designed star-schema data marts in PostgreSQL for Tableau dashboard consumption, trimming unnecessary columns and avoiding snowflake normalization on large fact tables to optimize query efficiency.
- Built and maintained an automated watchdog monitoring system for scheduled KNIME batch jobs, including stuck-job detection, automatic retry logic, and disk-space cleanup automation.

KPDN Data Pipeline Management (Apache NiFi)

- Redesigned production Apache NiFi pipeline architecture, consolidating unnecessary transformation nodes and resolving recurring stuck-pipeline failures, cutting processing duration from 30 to 20 minutes per run.

JDN Proof-of-Concept (Agentic AI Platform)

- Designed and built KNIME proof-of-concept workflows validating REST API integration, multi-source ETL/ELT, and classical ML/LLM-based sentiment analysis, forming the technical evidence base for the platform proposal; contributing to the technical architecture of a company-proposed 6-agent LangGraph supervisor system with a FastAPI services layer, currently at requirements gathering.

National Institute of Technology, Ube College, Japan | Researcher (Acoustic & Machine Learning) Intern

Oct 2023 - Jan 2024

- Conducted research on Detection of Respiratory Diseases from Auscultated Sounds Using VGG16 with Data Augmentation, an 8-class classification task using transfer learning and STFT-based audio feature extraction, achieving 75% accuracy with data augmentation, a 5-point improvement over the non-augmented baseline.
- Presented findings at the 2nd International Conference on Computer Graphics and Image Processing (CGIP 2024), Kyoto University.
- Extended the research into a CNN-LSTM architecture for a 5-class respiratory disease classification task, achieving 88% accuracy with data augmentation, presented in the internship report.

Northern Asia Resources | Business Analyst & Website Designer Intern

Sep 2021 - Feb 2022

- Developed an automated payroll system with QR code-based attendance tracking and leave application integration.
- Analyzed business processes and designed client websites; conducted system demos for government stakeholders.

KEY PROJECTS

TourGether (AI-Powered Travel Itinerary System, 2025)

- Built an intelligent travel planning system using Python and YOLOv11 for landmark/restaurant recognition from user images.
- Designed a hybrid RAG retrieval pipeline (FAISS + BM25) for personalized recommendations, built an agentic workflow using LangGraph and LangChain (tool-calling, Tavily web search, Supabase vector store) with OpenAI GPT API for itinerary generation; version-controlled on GitHub, deployed on Netlify.

Sentiment Analysis on NFT Project Authenticity, Final Year Project (2023-2024)

- Applied XGBoost, Random Forest, SVM, and MLP to classify genuine vs. fraudulent NFT projects from 13,626 tweets and 12,383 OpenSea transactions; XGBoost achieved 90.83% accuracy with cross-validation using an 80:20 split, visualized in an interactive Power BI dashboard.

EDUCATION

Universiti Teknologi MARA (UiTM) Shah Alam, Bachelor of Information System (Hons.),
Intelligent Systems Engineering, CGPA 3.81

2022 - 2024

Universiti Teknologi MARA (UiTM) Kedah, Diploma in Computer Science, CGPA 3.89

2019 - 2022

CERTIFICATIONS

- Certified Artificial Intelligence Proficient, SG Academy (M) Sdn Bhd (2026)
- AWS Certified Machine Learning Engineer - Associate, Amazon Web Services (2026)
- Basic Proficiency in KNIME Analytics Platform, KNIME (2026)
- Introduction to Data Engineering on Google Cloud, Google Cloud (2025)
- Google Data Analytics Professional Certificate, Google/Coursera (2024)

ACHIEVEMENTS & INVOLVEMENT

- Anugerah Inovasi (Innovation Award), Opensoft Technologies Sdn Bhd, for the innovative idea Local LLMs in KNIME Analytics Platform (January 2026)
- Vice Chancellor Award, Bachelor of Information System (Hons.), UiTM Shah Alam
- Vice Chancellor Award, Diploma in Computer Science, UiTM Kedah

LANGUAGES

Malay (Native), English (Proficient)

REFERENCE

Abdul Hakim bin Mohammad

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